Mitul Patil

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LinkedIn | GitHub | Portfolio

Professional Summary

Full Stack Developer with strong MERN stack and Node.js backend experience, hands-on experience with PostgreSQL, and working knowledge of Python/FastAPI. Experienced in production AI integration across RAG pipelines, LLM APIs, vector search, and autonomous agents. Built and deployed full-stack AI applications and shipped a live mobile application. Seeking entry-level software engineering roles building practical, AI-powered products.

Education

B.Tech in Computer Science & Engineering June 2022 – May 2026
Parul University, Vadodara, Gujarat
CGPA: 8.59

Technical Skills

Languages: JavaScript (ES6+), TypeScript, Java, Python, SQL
Frontend: React.js, Redux, React Native, HTML5, CSS3, Tailwind CSS, React Router v6, React Hook Form
Backend: Node.js, Express.js, REST API Design, Socket.IO, JWT Authentication, FastAPI, Pydantic, Depends(), Zod
Databases: PostgreSQL, pgvector, MongoDB, Supabase, Firebase
AI / LLM: Gemini SDK, RAG Pipelines, AI Agents, Embeddings, Vector Search, LangChain, LangGraph JS, Tavily API, SSE Streaming, Prompt Engineering
Tools & Deployment: Git, GitHub, Postman, Docker (Basics), CI/CD Pipelines (Basics), Vercel, Railway, Github Actions

Internship Experience

Full Stack Developer Intern – Exotic Infotech Jan 2026 – June 2026

- Translated Figma designs into **10+ pixel-perfect React components**, reducing UI review cycles by approximately **30%**
- Delivered **3 full-stack MERN modules** (authentication, dashboard, and orders), reducing manual admin effort by approximately **40%**
- Integrated **5+ REST APIs** using Axios and centralized error handling, reducing failed requests to under **2%**
- Optimized Redux state management, improving page-load responsiveness across **4 key user flows**

Projects

DocMind – RAG Document Intelligence App Live | GitHub
React, Node.js, Express, PostgreSQL, pgvector, Gemini SDK, Multer, JWT, Socket.IO, Supabase

- Built a **full RAG pipeline from scratch**, covering PDF parsing, semantic chunking, embedding generation, **pgvector cosine similarity search**, and **token-by-token LLM response streaming** via SSE to a React frontend
- Implemented **hybrid search** (semantic and keyword), source citations, and per-user **document isolation** using PostgreSQL **row-level security**, along with prompt-injection defenses
- Processed large PDFs asynchronously, logged **Gemini API token costs per user**, tested API endpoints using Postman, and deployed the application on Vercel, Railway, and Supabase

ResearchBot – Autonomous AI Research Agent GitHub
React, Node.js, Express, PostgreSQL, LangGraph JS, LangSmith, Gemini SDK, Tavily API

- Built an **autonomous research agent** using a **LangGraph JS state machine**, with planning, web search (Tavily), URL fetching, synthesis, and save nodes connected through **conditional edges**, plus human-in-the-loop approval and **LangSmith tracing** for workflow observability
- Reduced manual research effort through structured report generation using Gemini's **responseSchema** and **LangSmith evaluation** to monitor and improve agent response quality
- Implemented **real-time step streaming** to React via SSE, added prompt-injection defenses for AI agents, and optimized PostgreSQL queries using **EXPLAIN ANALYZE**

Greenlit – AI-Powered Freelancer Decision Assistant GitHub
React.js, Node.js, Express.js, PostgreSQL, LLM API, TipTap, JWT

- Automated proposal, scope, and risk document generation using an LLM, reducing creation time by approximately **70%**
- Engineered a **TipTap rich-text editor** with **8-second auto-save** and one-click PDF/DOCX export; built a **JWT-authenticated Discovery Portal** with a client Q&A workflow and shareable project workspace

Mr. Dhobi – On-Demand Laundry Mobile App Live (Play Store) | GitHub
React Native, TypeScript, Firebase, Razorpay

- Shipped a **live Android app on Google Play Store** with **OTP authentication**, push notifications, and **Razorpay** payment integration
- Implemented a **role-based admin panel** with **10+ operational modules** for managing orders, delivery, and user roles; the companion website achieved a **98/100 Lighthouse performance score**

Certifications

Full Stack Web Developer – Udemy (2025)